Kinetic energy and net work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net work to a change in kinetic energy.

SUCCESS CRITERIA - I can...

1. Calculate kinetic energy with speed squared.
2. Use net work rather than one force's work.
3. Connect the sign of work to a speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "kinetic energy" and "work" to explain your model or result. Sentence starters are optional.

kinetic energy: energy associated with motion

work: energy transferred by a force along a displacement

initial: at the beginning of the interval; subscript i

01 Bring back an earlier idea

Keep mass fixed and multiply speed by 3. By what factor does kinetic energy change?
Explain from the equation.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $K = \frac{1}{2}mv^2$; $W_{net} = K_f - K_i$.
i = initial; f = final. Positive net work increases kinetic energy.

Worked reasoning: A 2 kg cart at 2 m/s has K=4 J. Adding 5 J of net work gives final K=9 J, so final speed is 3 m/s.

Your turn: Draw before-and-after energy bars and label the transfer between them.

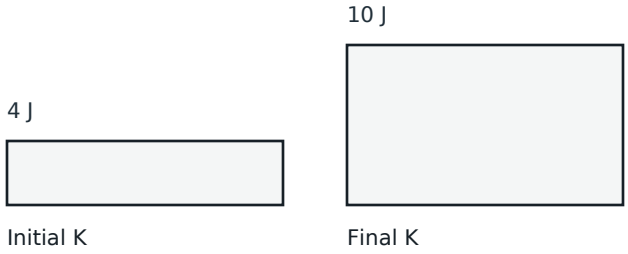
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

Energy bars show kinetic energy changing from 7 J to 10 J.
Find net work and describe the speed change for constant mass.

Sentence starter: I found … [unit]. This value means … in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg cart moves at 3 m/s.
Find $K=\frac{1}{2}\times2\times3^2$. If net work adds 7 J, find final kinetic energy.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A student adds the work of a pull but ignores friction. Why might their predicted speed be too high?

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

06 Choose a method

Two 4 kg carts latch together after colliding. Is this perfectly inelastic?
State what remains conserved if external impulse is negligible.

Word help: inelastic = a collision that transforms some kinetic energy to other forms

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

07 Exit check

K changes from 20 J to 8 J.
Find net work and state whether speed rises or falls.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Keep the cart's mass fixed.
Compare the energy needed to increase speed from 1 to 2 m/s with that from 2 to 3 m/s.
Use m=2 kg.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3